8. PERFORM ROOTKIT DETECTION AND REMOVAL USING RKHUNTER TOOL

DATE: 18/03/2026

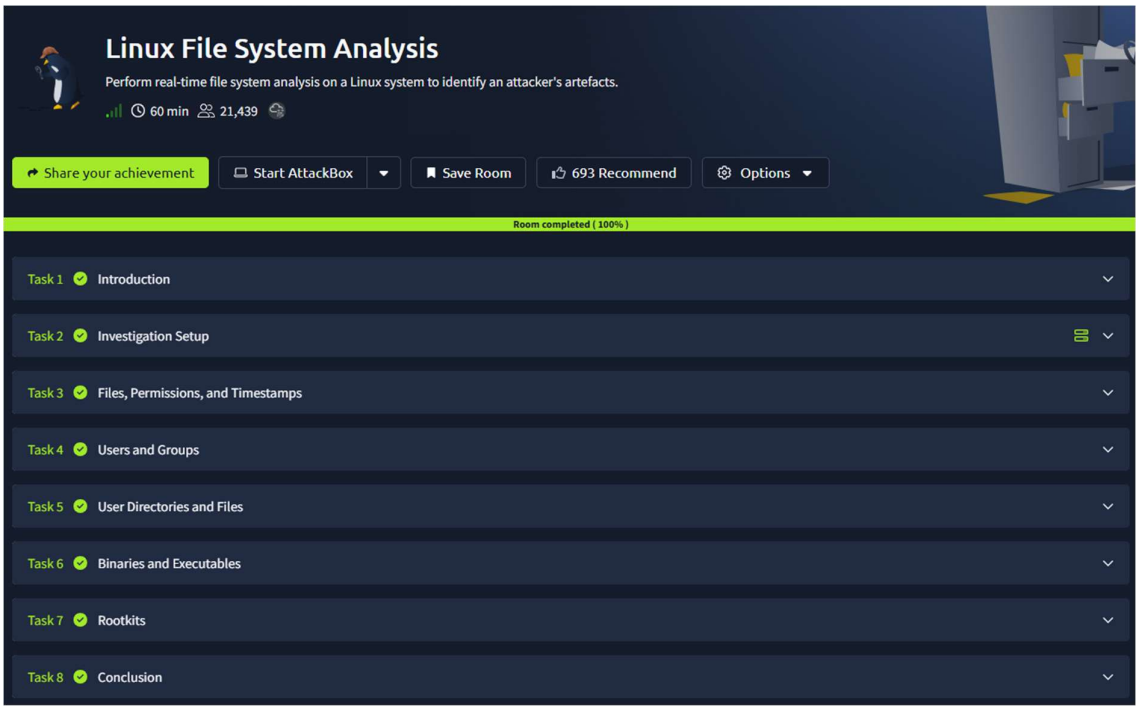
Aim:

To perform real-time file system analysis on a Linux system to identify an attacker's artefacts.

Procedure:

1. Log into TryHackMe and open the Linux File System Analysis room.
2. Explore the Linux file system structure and important directories.
3. Perform analysis to identify suspicious files and attacker artefacts.
4. Examine logs and metadata for signs of unauthorized activity.
5. Use basic commands and tools to investigate file changes.
6. Complete the tasks and answer the questions in the room.
7. Review findings and confirm completion.

Tasks:



TASK 1 - Introduction

TASK 2 – Investigation Setup

```
investigator@ip-10-49-181-214:~$  export PATH=/mnt/usb/bin:/mnt/usb/sbin
investigator@ip-10-49-181-214:~$ export LD_LIBRARY_PATH=/mnt/usb/lib:/mnt/usb/lib64
investigator@ip-10-49-181-214:~$ check-env
THM{5514ec4f1ce82f63867806d3cd95dbd8}
```

Answer the questions below

After updating the `PATH` and `LD_LIBRARY_PATH` environment variables, run the command `check-env`. What is the flag that is returned in the output?

THM{5514ec4f1ce82f63867806d3cd95dbd8} ✓ Correct Answer

TASK 3 – Files, Permissions and Timestamps

```
investigator@ip-10-49-181-214:/var/tmp$ find / -user bob -type f -cmin -1
find: '/sys/kernel/tracing': Permission denied
find: '/sys/kernel/debug': Permission denied
find: '/sys/fs/pstore': Permission denied
find: '/sys/fs/bpf': Permission denied
find: '/proc/tty/driver': Permission denied
find: '/proc/1/task/1/fd': Permission denied
find: '/proc/1/task/1/fdinfo': Permission denied
find: '/proc/1/task/1/ns': Permission denied
find: '/proc/1/fd': Permission denied
find: '/proc/1/map_files': Permission denied
find: '/proc/1/fdinfo': Permission denied
find: '/proc/1/ns': Permission denied
find: '/proc/2/task/2/fd': Permission denied
```

```
find: '/var/tmp/systemd-private-f37682276e6d4bc483ae1882bb077c40-ModemManager.service-cgKgQi': Permission denied
find: '/var/tmp/systemd-private-f37682276e6d4bc483ae1882bb077c40-systemd-timesyncd.service-1vhpEh': Permission de
nied
/var/tmp/findme.txt
find: '/var/tmp/systemd-private-f37682276e6d4bc483ae1882bb077c40-apache2.service-B4f4Wg': Permission denied
find: '/var/tmp/systemd-private-f37682276e6d4bc483ae1882bb077c40-systemd-resolved.service-aq0tig': Permission den
ied
find: '/var/snap/lxd/common/lxd': Permission denied
```

```
investigator@ip-10-49-181-214:/var/tmp$ cat /var/tmp/findme.txt
THM{0b1313afd2136ca0faafb2daa2b430f3}
```

```
investigator@ip-10-49-181-214:/var/www/html/assets$ exiftool reverse.elf
ExifTool Version Number         : 11.88
File Name                       : reverse.elf
Directory                       : .
File Size                       : 250 bytes
File Modification Date/Time     : 2024:02:13 00:26:28+00:00
File Access Date/Time           : 2024:02:13 00:32:59+00:00
File Inode Change Date/Time     : 2024:02:13 00:34:50+00:00
File Permissions                : rwxr-xr-x
File Type                       : ELF executable
File Type Extension             :
MIME Type                       : application/octet-stream
CPU Architecture                : 64 bit
CPU Byte Order                  : Little endian
Object File Type                : Executable file
CPU Type                        : AMD x86-64
```

```
investigator@ip-10-49-181-214:/etc$ stat hosts
  File: hosts
  Size: 221             Blocks: 8          IO Block: 4096   regular file
Device: ca01h/51713d    Inode: 49          Links: 1
Access: (0644/-rw-r--r--)  Uid: (    0/    root)   Gid: (    0/    root)
Access: 2026-04-20 08:30:32.844000000 +0000
Modify: 2020-10-26 21:10:44.000000000 +0000
Change: 2020-10-26 23:32:25.957900650 +0000
 Birth: -
```

To practice your skills with the `find` command, locate all the files that the user **bob** created in the past 1 minute. Once found, review its contents. What is the flag you receive?

THM{0b1313afd2136ca0faafb2daa2b430f3} ✓ Correct Answer

Extract the metadata from the `reverse.elf` file. What is the file's MIME type?

application/octet-stream ✓ Correct Answer

Run the `stat` command against the `/etc/hosts` file on the compromised web server. What is the full **Modify Timestamp (mtime)** value?

2020-10-26 21:10:44.000000000 +0000 ✓ Correct Answer

TASK 4 – Users and Groups

```
investigator@ip-10-49-181-214:~$ cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/var/run/ircd:/usr/sbin/nologin
gnats:x:41:41:Gnats Bug-Reporting System (admin):/var/lib/gnats:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:100:102:systemd Network Management,,,:/run/systemd:/usr/sbin/nologin
systemd-resolve:x:101:103:systemd Resolver,,,:/run/systemd:/usr/sbin/nologin
systemd-timesync:x:102:104:systemd Time Synchronization,,,:/run/systemd:/usr/sbin/nologin
messagebus:x:103:106::/nonexistent:/usr/sbin/nologin
syslog:x:104:110::/home/syslog:/usr/sbin/nologin
_apt:x:105:65534::/nonexistent:/usr/sbin/nologin
tss:x:106:111:TPM software stack,,,:/var/lib/tpm:/bin/false
uuidd:x:107:112::/run/uuidd:/usr/sbin/nologin
tcpdump:x:108:113::/nonexistent:/usr/sbin/nologin
sshd:x:109:65534::/run/sshd:/usr/sbin/nologin
landscape:x:110:115::/var/lib/landscape:/usr/sbin/nologin
pollinate:x:111:1::/var/cache/pollinate:/bin/false
ec2-instance-connect:x:112:65534::/nonexistent:/usr/sbin/nologin
systemd-coredump:x:999:999:systemd Core Dumper:/:/usr/sbin/nologin
ubuntu:x:1000:1000:Ubuntu:/home/ubuntu:/bin/bash
lxd:x:998:100::/var/snap/lxd/common/lxd:/bin/false
bob:x:1001:1001::/home/bob:/bin/bash
jane:x:1002:1002:Jane Walkers,103,9399499494,2029384958:/home/jane:/bin/bash
investigator:x:1003:1003:Investigator,1,1,1,1:/home/investigator:/bin/bash
postfix:x:113:120::/var/spool/postfix:/usr/sbin/nologin
b4ckd00r3d:x:0:1004::/home/b4ckd00r3d:/bin/sh
```

```
investigator@ip-10-49-181-214:~$ cat /etc/group | grep 46
plugdev:x:46:ubuntu,investigator
```

```
investigator@ip-10-49-181-214:~$ sudo cat /etc/sudoers |grep jane
[sudo] password for investigator:
jane ALL=(ALL) /usr/bin/pstree
```

Answer the questions below

Investigate the user accounts on the system. What is the name of the backdoor account that the attacker created?

b4ckd00r3d ✓ Correct Answer

What is the name of the group with the group ID of **46**?

plugdev ✓ Correct Answer

View the `/etc/sudoers` file on the compromised system. What is the full path of the binary that Jane can run as sudo?

/usr/bin/pstree ✓ Correct Answer

TASK 5 – User Diresctories and Files

```
investigator@ip-10-49-181-214:/home/jane$ sudo cat .bash_history
whoami
groups
cd ~
ls -al
find / -perm -u=s -type f 2>/dev/null
/usr/bin/python3.8 -c 'import os; os.execl("/bin/sh", "sh", "-p", "-c", "cp /bin/bash /var/tmp/bash && chown r
oot:root /var/tmp/bash && chmod +s /var/tmp/bash")'
ls -al /var/tmp
exit
useradd -o -u 0 b4ckd00r3d
exit
THM{f38279ab9c6af1215815e5f7bbad891b}
```

```
investigator@ip-10-49-181-214:/home/bob$ ls -la
total 36
drwxr-xr-x 4 bob  bob  4096 Feb 12  2024 .
drwxr-xr-x 6 root root 4096 Feb 12  2024 ..
-rw-r--r-- 1 bob  bob   220 Feb 12  2024 .bash_logout
-rw-r--r-- 1 bob  bob  3771 Feb 12  2024 .bashrc
drwx------ 2 bob  bob  4096 Feb 12  2024 .cache
-rw-rw-r-- 1 bob  bob     0 Feb 12  2024 .hidden1
-rw-rw-r-- 1 bob  bob     0 Feb 12  2024 .hidden10
-rw-rw-r-- 1 bob  bob     0 Feb 12  2024 .hidden11
-rw-rw-r-- 1 bob  bob     0 Feb 12  2024 .hidden12
-rw-rw-r-- 1 bob  bob     0 Feb 12  2024 .hidden13
-rw-rw-r-- 1 bob  bob     0 Feb 12  2024 .hidden14
-rw-rw-r-- 1 bob  bob     0 Feb 12  2024 .hidden15
-rw-rw-r-- 1 bob  bob     0 Feb 12  2024 .hidden16
-rw-rw-r-- 1 bob  bob     0 Feb 12  2024 .hidden17
-rw-rw-r-- 1 bob  bob     0 Feb 12  2024 .hidden18
-rw-rw-r-- 1 bob  bob     0 Feb 12  2024 .hidden19
-rw-rw-r-- 1 bob  bob     0 Feb 12  2024 .hidden2
-rw-rw-r-- 1 bob  bob     0 Feb 12  2024 .hidden20
-rw-rw-r-- 1 bob  bob     0 Feb 12  2024 .hidden21
-rw-rw-r-- 1 bob  bob     0 Feb 12  2024 .hidden22
-rw-rw-r-- 1 bob  bob     0 Feb 12  2024 .hidden23
-rw-rw-r-- 1 bob  bob     0 Feb 12  2024 .hidden24
-rw-rw-r-- 1 bob  bob     0 Feb 12  2024 .hidden25
-rw-rw-r-- 1 bob  bob     0 Feb 12  2024 .hidden26
-rw-rw-r-- 1 bob  bob     0 Feb 12  2024 .hidden27
-rw-rw-r-- 1 bob  bob     0 Feb 12  2024 .hidden28
-rw-rw-r-- 1 bob  bob     0 Feb 12  2024 .hidden29
-rw-rw-r-- 1 bob  bob     0 Feb 12  2024 .hidden3
-rw-rw-r-- 1 bob  bob     0 Feb 12  2024 .hidden30
-rw-rw-r-- 1 bob  bob     0 Feb 12  2024 .hidden31
-rw-rw-r-- 1 bob  bob     0 Feb 12  2024 .hidden32
-rw-rw-r-- 1 bob  bob     0 Feb 12  2024 .hidden33
-rw-rw-r-- 1 bob  bob    38 Feb 12  2024 .hidden34
```

```
investigator@ip-10-49-181-214:/home/bob$ cat .hidden34
THM{6ed90e00e4fb7945bead8cd59e9fcd7f}
```

```
investigator@ip-10-49-181-214:/home/jane$ cd .ssh
investigator@ip-10-49-181-214:/home/jane/.ssh$ ls -la
total 20
drwxr-xr-x 2 jane jane 4096 Feb 12  2024 .
drwxr-xr-x 4 jane jane 4096 Feb 13  2024 ..
-rw-rw-rw- 1 jane jane 1136 Feb 13  2024 authorized_keys
-rw------- 1 jane jane 3389 Feb 12  2024 id_rsa
-rw-r--r-- 1 jane jane  746 Feb 12  2024 id_rsa.pub
investigator@ip-10-49-181-214:/home/jane/.ssh$ stat authorized_keys
  File: authorized_keys
  Size: 1136            Blocks: 8          IO Block: 4096   regular file
Device: ca01h/51713d    Inode: 257561      Links: 1
Access: (0666/-rw-rw-rw-)  Uid: ( 1002/    jane)   Gid: ( 1002/    jane)
Access: 2024-02-13 00:34:53.692530853 +0000
Modify: 2024-02-13 00:34:16.005897449 +0000
Change: 2024-02-13 00:34:16.005897449 +0000
 Birth: -
```

Answer the questions below

View Jane's `.bash_history` file. What flag do you see in the output?

THM{f38279ab9c6af1215815e5f7bbad891b} ✓ Correct Answer

What is the hidden flag in Bob's home directory?

THM{6ed90e00e4fb7945bead8cd59e9fcd7f} ✓ Correct Answer

Run the `stat` command on Jane's `authorized_keys` file. What is the full timestamp of the most recent modification?

2024-02-13 00:34:16.005897449 +0000 ✓ Correct Answer

TASK 6 – Binaries and Executables

```
investigator@ip-10-49-181-214:/etc$ sudo debsums -c -e
/etc/sudoers
investigator@ip-10-49-181-214:/etc$ md5sum /var/tmp/bash
7063c3930affe123baecd3b340f1ad2c  /var/tmp/bash
```

Answer the questions below

Run the `debsums` utility on the compromised host to check only configuration files. Which file came back as altered?

/etc/sudoers ✓ Correct Answer

What is the `md5sum` of the binary that the attacker created to escalate privileges to root?

7063c3930affe123baecd3b340f1ad2c ✓ Correct Answer

TASK 7 - Rootkits

```
Searching for common ssh-scanners default files...          nothing found
Searching for Linux/Ebury - Operation Windigo ssh...        nothing found
Searching for 64-bit Linux Rootkit ...                      nothing found
Searching for 64-bit Linux Rootkit modules...               nothing found
Searching for Mumblehard Linux ...                          * * * * * /var/tmp/findme.sh
```

```
investigator@ip-10-49-181-214:/$ sudo rkhunter --check --sk --rwo | grep UID
Warning: Account 'b4ckd00r3d' is root equivalent (UID = 0)
```

```
investigator@ip-10-49-181-214:/$ sudo chkrootkit
ROOTDIR is `/'
Checking `amd'...                                           not found
Checking `basename'...                                      not infected
Checking `biff'...                                          not found
Checking `chfn'...                                          not infected
Checking `chsh'...                                          not infected
Checking `cron'...                                          not infected
Checking `crontab'...                                       not infected
Checking `date'...                                          not infected
Checking `du'...                                            not infected
Checking `dirname'...                                       not infected
Checking `echo'...                                          not infected
Checking `egrep'...                                         not infected
Checking `env'...                                           not infected
Checking `find'...                                          not infected
Checking `fingerd'...                                       not found
Checking `gpm'...                                           not found
Checking `grep'...                                          not infected
Checking `hdparm'...                                        not infected
Checking `su'...                                            not infected
Checking `ifconfig'...                                      not infected
Checking `inetd'...                                         not infected
```

Answer the questions below

Run *chkrootkit* on the affected system. What is the full path of the `.sh` file that was detected?

/var/tmp/findme.sh — ✓ Correct Answer

Run *rkhunter* on the affected system. What is the result of the `(UID 0) accounts` check?

Warning — ✓ Correct Answer

TASK 8 - Conclusion

Answer the questions below

Click and continue learning!

No answer needed — ✓ Correct Answer

Result:

Thus the TryHackMe room Linux File System Analysis was completed successfully.